Phishing Attack Detection Using Machine Learning

Ahmad Al-khateeb . Shahed Odeh . Faris Salah

Abstract

A phishing attack is a type of cyber-attack where the attacker attempts to trick individuals into revealing sensitive information, such as login credentials is a serious threat both to consumers and institutions doing business on the web. The increase in the rate of phishing attacks is a concerning trend in the world of cybersecurity. Not only that, but it has become more and more complex and dangerous which is difficult to detect. Therefore, there is a great need to use intelligent algorithms to solve this problem.one of the best ways for detecting these malicious acting is to develop machine learning automated learning models will be trained to analyze patterns, behaviors, and characteristics of phishing attacks to detect the threat as soon as it occurs. By continuously learning from new databases and adapting to emerging threats, machine learning-based phishing detection systems enhance an organization's ability to protect against fraudulent email and other forms of phishing. Our proposed topic performs better compared to other phishing related topics. This declares that our approach can be used on the web for real-time threat detection using phishing URLs detection.

1 Introduction

Phishing Attacks are the most common ways of attack in the digital world these days. Any method of communication can be used to target an individual to trick them into leaking confidential data in a fake environment, which can later be used to harm the sole victim or even an entire business depending on the attacker's intent and the type of data leaked. Phishing attacks, while dangerous, can be avoided by simply creating awareness and developing habits of staying alert and continuously being on the lookout when surfing through the internet and only clicking links after verifying if the source of the links is trustworthy.

There are also tools such as browser extensions that notify users when they have entered their credentials on a fake site, possibly having their credentials transferred to a user with malicious intent.

Other tools can also allow networks to lock down everything and allow access to whitelisted sites to provide extra security while compromising some convenience on the user side [6]. In a related study, three main reasons have been stated behind users falling into traps of phishing attack schemes:

- ✓ Lack of knowledge about URLs.
- ✓ Lack of knowledge about trusted websites.
- ✓ Lack of visibility of full web addresses due to the redirection or hidden URLs.

A recent Microsoft security intelligence (volume 24) report found that phishing attacks were on the top of the discovered web attacks of 2018, and it is expected to continue increasing. The major challenge when detecting phishing attacks lies in discovering the techniques utilized. Phishers continuously enhance their strategies

and can create web pages that are able to protect themselves against many forms of detection. Accordingly, developing robust, effective, and up-to-date phishing detection methods is necessary to oppose the adaptive techniques employed by the phishers.

The blacklist: approach maintains a list of Suspicious or malicious URLs that are collected using different approaches like Google safe browsing, Phish Tank, and user voting. So, when a web page is initiated, the browser searches the blacklist for it and alerts the user if the webpage was found. Finally, the blacklist can be stored on the user's machine or on a server [7]. Blacklists are often used to classify websites as malicious or legitimate.

The content-based: approach deploys an in-depth analysis of the page's content. Building classifiers and extracting features from page contents and third-party services such as search engines and DNS servers. Yet, these methods are ineffective because of a massive number of training features and the reliance on third-party servers which assault user's privacy by uncovering their browsing history.

The Heuristic-Based: Approach the detection technique is based on employing various discriminative features extracted by understanding and analyzing the structure of phishing web pages.

The aim of this paper is to present a study of existing methods used in the detection of phishing web pages that employed machine learning algorithms and focus on the

most common feature selection methods that are used for dealing with various problems and enhance the performance and effectiveness of phishing datasets. Moreover, we will apply feature selection to an existing novel phishing data set to enhance the effectiveness of the data set and decrease the time taken to build the models.

2 Literature Review

This section of the literature Review eventually reveals some facts based on thoughtful analysis of many authors' works as follows:

Sahingoz, O. K., Buber, E., Demir, O., & Diri, B. “Machine Learning-Based Phishing Detection from URLs,” Expert Systems with Applications, 117:345-357, January 2019: The dataset used is self-constructed. Where phishing websites belong to PhishTank and legitimate URLS are from Yandex Search API. The main purpose was to detect the word which is similar to brand names, to detect keywords, the words, which are formed from random characters. Various classification algorithms such as Naive Bayes, Random Forest, kNN(n=3), Adaboost, K-star, SMO, and Decision Tree including some feature extraction types such as NLP-based features, Word Vectors and Hybrid are used. This system gets high accuracy throughout the test [1].

J. James, Sandhya L. and C. Thomas, “Detection of phishing URLs using machine learning techniques,” International Conference on Control Communication and Computing (ICCC), December 2013: The system proposed used a method based on lexical features, host properties, and properties related to the page for the detection of phishing websites. For getting a proper understanding of the pattern

of URLs, various data mining algorithms are used. The classification algorithms such as Naïve Bayes, J48 Decision Tree, K-NN, and SVM were considered for the detection of phishing websites. Decision Tree had better accuracy of 91.08% compared to other algorithms. So, Tree-based classifiers are best suited for phishing URL classification [2].

Pradeepthi, K. V., & Kannan, A. “Performance study of classification techniques for phishing URL detection,” Sixth International Conference on Advanced Computing (IcoAC), December 2014.: The system recognizes Phishing URLs, by examining the URL structure without attending the Phishing URL using classification algorithms. The data collected is first passed through the training state where it undergoes feature selection and classification. The dataset used here contains 4500 URL records, on which classification is performed. Out of which 2500 URLs are genuine and the other 2000 are the phishing ones. 2500 URLs were collected from the DMOZ repository. The 2000 phishing URLs have been picked from PHISHTANK. Data claswereication after extraction of the relevant features was performed by Naive Bayes, Random Forest, Random Tree, Multi-layer Perceptron, C-RT, J 48 Tree, LMT, C 4.5, ID 3, and K-Nearest Neighbour. The Random Forest Algorithm had the highest classification accuracy [3].

Dipayan Sinha, Dr. Minal Moharir, Prof. Anitha Sandeep, “Phishing Website URL Detection using Machine Learning,” International Journal of Advanced Science and Technology, 29(3):2495-2504, 2020: Detection of phishing websites is performed by using machine learning techniques like Logistic Regression, Decision tree, Random Forest, Adaboost, Gradient Boosting, Gaussian NB, and Fuzzy pattern tree classifier. Data collection involves phishing and legitimate websites. Extracting useful features

has two steps: URL-based involves IP Address, '@' symbol in URL, dashes in URL, long URL, presence of unusual number, dot count, sub-domains in URL, etc. Domain based includes Page Rank of the website, age of the Domain, and Validity of the Website. The dataset is split into training and testing set in the ratio 80:20. The Random Forest algorithm shows 96% precision and recalls along with the highest F1 score of 95% [4].

R. Kiruthiga, D. Akila, “Phishing Websites Detection Using Machine Learning,” International Journal of Recent Technology and Engineering (IJRTE), 8(2S11):11-114, September 2019:2019 A total of 15 research papers have been studied in this research paper. In this research, one method was discussed which uses five different algorithms that are Decision Tree, Generalized Linear Model, Gradient Boosting, Generalized Additive Model, and Random Forest. On comparing the results, the Random Forest algorithm had the highest accuracy of 98.4%, 98.59% recall, and precision of 97.70%. The dataset used is from the UCI machine learning repository [5].

3 Methodology

3.1 Phishing Website Dataset

The provided dataset includes 11430 URLs with 87 extracted features. The dataset is designed to be used as a benchmark for machine learning-based phishing detection systems. Features are from three different classes: 56 are extracted from the structure and syntax of URLs, 24 are extracted from the content of their

correspondent pages, and 7 are extracted by querying external services. The dataset is balanced, it contains exactly 50% phishing and 50% legitimate URLs.

features taken out from phishing web pages and legitimate web pages, Phishing webpages are collected from Phish-Tank and Open-Phish, while legitimate web pages are collected from Common Crawl. Features in this dataset are classified into three groups, which are Address bar-based, Abnormal-based, and HTML/JavaScript-based features. Address bar-based are the features in the URL of the web page like URL's length and port number, abnormal-based are features of abnormal actions on the web page like downloading objects from external domains, and HTML/JavaScript-based are features of HTML and JavaScript methods placed in the source code of the web page [8]. In this work, we chose this dataset because it is the most recent dataset in this field.

3.2 Machine Learning Techniques

Different experiments have been done on different machine learning classifiers such as Bayes net, Naive Bayes, J48, Logistic, Random Forest, Bagging, and Multilayer perceptron. Then we chose three algorithms which obtained the best accuracy rates and the most used classifiers based on the literature, which are J48, Random Forest and Multilayer perceptron.

1. **J48 :**

J48 algorithm is one of the most widely used machine learning algorithms to examine the data categorically and continuously. The C4.5 algorithm (J48) is mostly used among many fields for classifying data for example interpreting the clinical data for the diagnosis of coronary heart disease, classifying E-governance data, and many more.J48 Classification and its Decision Tree The C4.5 algorithm is a classification algorithm which produces decision trees based on information theory. The J48 implementation of the C4.5 algorithm has many additional features including accounting for missing values, decision trees pruning, continuous attribute value ranges, derivation of rules, etc. In the WEKA data mining tool, J48 is an open-source Java implementation of the C4.5 algorithm. J48 allows classification via either decision trees or rules generated from them [9],[10].

This algorithm builds decision trees based on a set of training data in the same way the ID3 algorithm does, by using the concept of information entropy. The training data is a set S={s1, s2, …} of already classified samples. Each sample si consists of a p-dimensional vector (x1, i, x2, i, …, xp, i) where the xj represents the attribute values or features of the corresponding sample, as well as the class in which the sample falls. To gain the highest classification accuracy, the best attribute to split on is the attribute with the greatest information.

At each node of the tree, the C4.5 algorithm chooses the attribute of the data that most effectively splits its set of samples into subsets, enriched in one class or the other. The splitting criterion is the normalized information gain, which is calculated from the difference in entropy. The attribute with the highest normalized information gain is chosen to make the decision. The C4.5 algorithm then recurses

on the partitioned sub lists utilizing a divide-and-conquer approach and creates a decision tree based on the greedy algorithm [11].

The Decision Tree

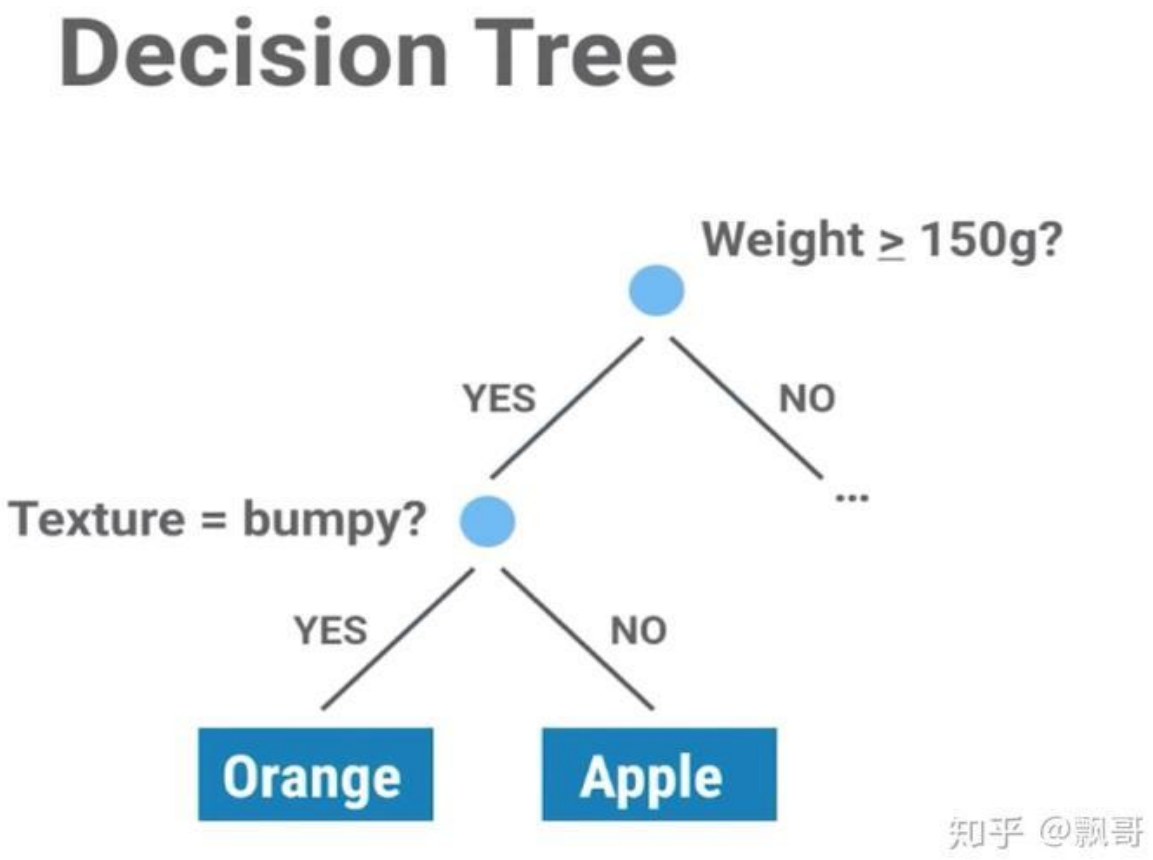
The internal nodes of this decision tree denote the different attributes, and the branches between the nodes tell us the possible values that these attributes can have in the observed samples, and the terminal nodes tell us the final value (classification) of the dependent variable. The predicted attribute is the dependent variable and the other attributes in the tree are the independent variables in the dataset.

When building a tree, J48 ignores the missing values; that item's value can be predicted from the attribute values for the other records.

The basic idea is to divide the data into ranges based on the attribute values found in the training sample.

Some basic use cases in this algorithm are:

- ✓ This algorithm creates a list for all samples.
- ✓ It evaluates features to assess any information gain, if no gain is possible, then this algorithm creates a node higher up in the tree using the expected value of the class.
- ✓ If this algorithm encounters an unknown class that it has not seen so far, then it creates a node higher up in the tree using the expected value.



2. Random Forest

Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique. It can be used for both Classification and Regression problems in ML. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.

As the name suggests, "Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset." Instead of relying on one decision

tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output.

The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.

The below diagram explains the working of the Random Forest algorithm:

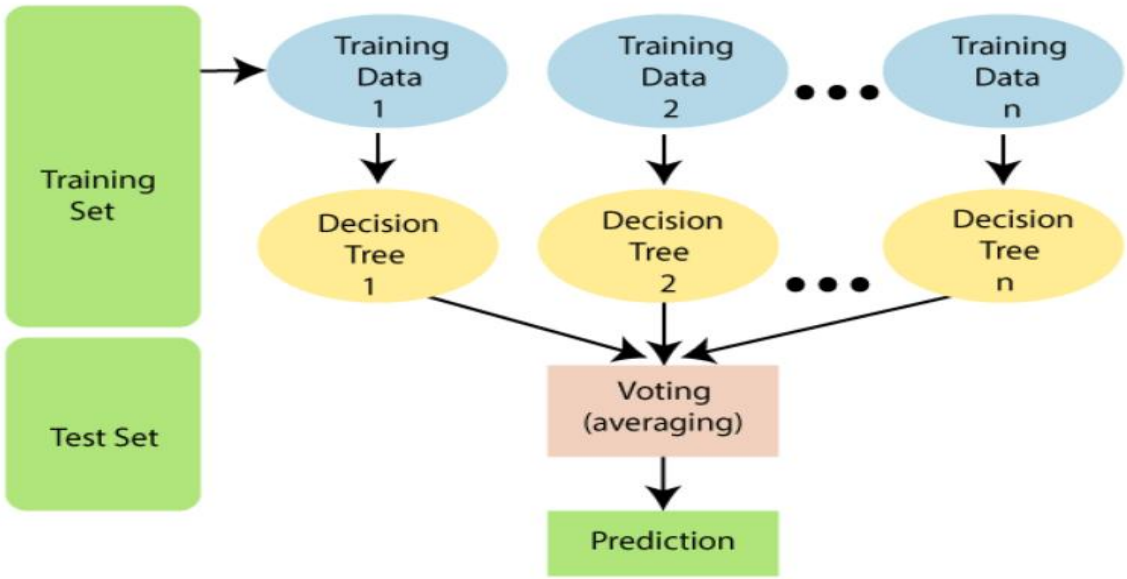


Fig. 2. Random Forest Structure

Why use Random Forest?

Below are some points that explain why we should use the Random Forest algorithm:

- ✓ It takes less training time as compared to other algorithms.
- ✓ It predicts output with high accuracy, even for the large dataset it runs efficiently.

- ✓ It can also maintain accuracy when a large proportion of data is missing [12].

3. Multilayer Perceptron Algorithm

A Multilayer Perceptron (MLP) is the most popular and frequently used artificial neural network. Like a neural network, MLP consists of multi interconnected components. They are constructed of three different layers which are an input layer, hidden layer, and output layer each has its own functionality (Fig. 3), an input layer is used to obtain the signal, an output layer turns out a decision about the input, and there is at least one hidden layer that is the computational engine of the MLP. It is usually utilized to supervise learning problems: it is trained on a group of input-output pairs and learns the correlation and dependencies among them [13].

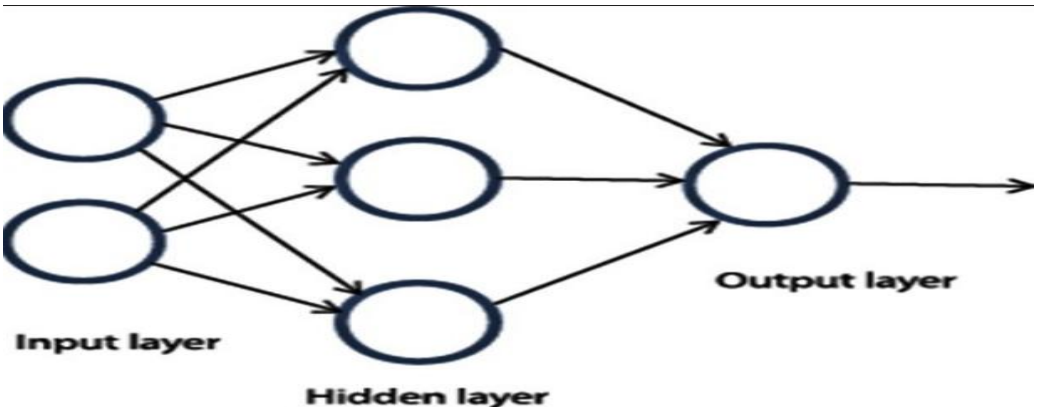


Fig. 3. Multilayer Perceptron Structure

4 Feature Selection

Feature selection is employed to decrease the size of the data to enhance the model's performance and reduce the computation time. Simply, the feature selection keeps the most important fields and eliminates unimportant ones.

However, it also gives useful and robust results. In this work, different feature selection methods will be utilized to enhance the phishing detection method by increasing the accuracy rate and decreasing the time taken to build the model.

Feature Selection Methods

Feature selection methods are classified based on the evaluation criteria into three categories, which are filters, wrappers, and hybrid methods.

In filter methods, the features are chosen based on the performance measure with the independence of the used data modeling algorithm or any utilized predictor. Then, after picking out the best features, the modeling algorithm can employ them.

In wrapper methods, the feature subsets are considered based on the quality of the performance of modeling the algorithm.

This method is significantly slower than the filter method in finding Excel lent feature subsets because it depends on the modeling algorithm. Whereas the wrapper method is more efficient in acquiring feature subsets than the filter method because the subsets are assessed using an actual modeling algorithm.

In the hybrid method, the best characteristics of filter and wrapper methods are combined. Primary, a filter method is employed to decrease the feature space. used to asset the best subsets, Hybrid methods obtained high accuracy and high efficiency rates [14].

The aim of this study is to assess different feature selection techniques in terms of accuracy and computational execution. Out of the overall 48 features used in phishing detection, some features will be optional in detecting phishing web pages.

Therefore, the essential features are taken away from the original dataset that is particularly effective in phishing detection, which will be debated in the results section.

Different experiments had been done on different filter methods of feature selection techniques such as InfoGain, ReliefF, PCA, and attribute. However, InfoGain and ReliefF had been chosen in our work because they attain the best accuracy rates than the remnant techniques.

InfoGain: It shows the significance of the features and determines which one of them is the most helpful for distinguishing among the classes. The value of InfoGain is calculated in the training data set. It is used in decision tree algorithms because it can help in deciding the best split; which high value indicates that split is excellent and low value indicates that the split is not good enough. The equation (1) used to estimate the value of an attribute by calculating the information gain according to the class [10].

ReliefF: As a filter-based feature selection method, Relief used to evaluate the quality of every feature according to the context of other features and the relevance of the feature to a given target notion [15]. The produced value of the algorithm is between - 1 and 1 for every feature in addition to positive numbers designating more significance or weighted attributes, the weight of an attribute is reduplicative upgraded, and it has a probabilistic description, the fundamental principle of relief is that important attributes are equivalent to instances of the same class.InfoGain(Class,Attribute) = H(Class)−H(Class|Attribute).

5 Model Evaluation

To evaluate the models, there are many assessment tools, But we intend to evaluate our model using the accuracy equation because the utilized dataset is a Binary and Balanced data set. So, calculating the accuracy rates will be enough, efficient, and accurate, To apply the accuracy formula, we should mention that there are two kinds of classification methods in accordance with the number of classes which are binary classification and multi-class classification, Where in binary classification there are only two classes whereas in multi-class classification the number of classes is more than two, In binary classes (Fig. 4), assume we have two classes, P for the positive class and N for the negative class [16].

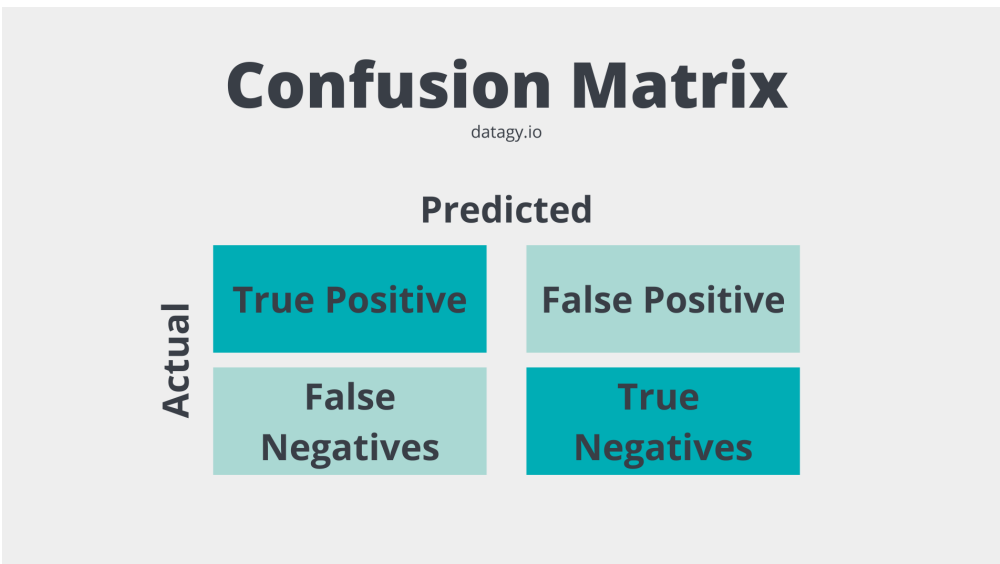


Fig. 4. An Example of The 2 X 2 Confusion Matrix

- ✓ **True Positive (TP): the true prediction rate of the positive samples. The predicted value is positive, and the actual value is also positive.**
- ✓ **False Positive (FP): negative value incorrectly classified as positive.**

- ✓ **True Negative (TN): the true prediction of negative samples. The predicted value is negative, and the actual value is also negative.**
- ✓ **False Negative (FN): positive value incorrectly classified as negative.**

Accuracy refers to the ratio of correctly classified instances. It is the most used evaluation metric for the performance of binary classification problems. Also, it is determining the accuracy.

$$Accuracy = \frac{\sum true\ positive + \sum true\ negative}{\sum total\ population}$$

6 Experiments and Results

In this study, the dataset mentioned in section 3 was employed, which contains 48 different features. For analysis and comparison between used classifiers, Weka has been utilized. Weka is a set of machine-learning algorithms used for different data mining functions such as data preparation, classification, regression, clustering, association rules mining and visualization.

Two different feature selection algorithms have been used in this study: InfoGain and ReliefF.

The details of the top 15 extracted features from both algorithms are described

Method	**Top 15 Features**
InfoGain	27, 28, 48, 34, 14, 35, 47, 5, 39, 1, 30, 3, 22, 25, 23
ReleifF	48, 30, 35, 34, 47, 40, 27, 39, 28, 29, 45, 31, 3, 26, 14

For the experiments, the 10-fold cross-validation technique is utilized in testing the models because it minimizes the estimation variance. By using this technique, the training dataset should be divided into 10 subsets, and then each of these subsets must be tested in the remaining nine subsets. Every test subset is employed once a time in all 10 repetitions. Tables 1,2 and 3 show the performance of the three selected algorithms (J48, RF, and MLP) using infoGain and reliefF feature selection methods with the top 5, top 10, and top 15 features.

Tables 1 The Performance of J48 Algorithm

Algorithm	Accuracy	Taken Time (seconds)
J48	97.31	1.2
J48+infogain+top5	95.31	0.12
J48+infogain+top10	96.17	0.21
J48+infogain+top15	96.96	0.35
J48+ reliefF +top5	89.59	0.08
J48+ reliefF +top10	97.08	0.16
J48+ reliefF +top15	97.28	0.29

Table 2 The Performance Of Random Forest (RF) Algorithm

Algorithm	Accuracy	Taken Time (seconds)
Random Forest (RF)	98.37	4.18
RF+infogain+top5	95.96	2.24
RF+infogain+top10	96.87	2.87
RF+infogain+top15	97.91	2.68
RF+ reliefF +top5	89.75	1.25
RF+ reliefF +top10	97.7	2.29
RF+ reliefF +top15	97.87	2.48

Table 3 The Performance of Multilayer Perceptron (MLP) Algorithm

Algorithm	Accuracy	Taken Time (seconds)
MLP	96.59	117.79
MLP +infogain+top5	91.89	6.04
MLP +infogain+top10	93.45	12.02
MLP +infogain+top15	95.74	18.92
MLP + reliefF +top5	88.22	4.75
MLP + reliefF +top10	95.63	9.74
MLP + reliefF +top15	96.19	14.93

7 Conclusion

These days there are an enormous number of web pages, and phishing web pages take a significant part of them. Phishing web pages are trying to lure users to get the benefits from them. This paper proposed a method of phishing detection using machine learning algorithms and employing a dataset of legitimate web pages and phishing ones. The best results are acquired by utilizing feature selection tools. The best accuracy they achieved was 91.08 using Decision Tree, but our teams were able to reach an accuracy of 98.37 within 4.18 seconds using Random Forest (RF) , but it is a long time, so we repeated the experiment and obtained an accuracy of 95.31 in a record time of 0.12 using J48+infogain+top5. It then saves the phishing information in Blacklist for faster and more accurate re-access the next time.

References

[1] Sahingoz, O. K., Buber, E., Demir, O., & Diri, B. “Machine Learning-Based Phishing Detection from URLs,” Expert Systems with Applications, vol. 117, pp. 345-357, January 2019.

[2] J. James, Sandhya L. and C. Thomas, “Detection of phishing URLs using machine learning techniques,” International Conference on Control Communication and Computing (ICCC), December 2013.

[3] Pradeepthi, K. V., & Kannan, A. “Performance study of

classification techniques for phishing URL detection,”

Sixth International Conference on Advanced Computing

(IcoAC), December 2014.

[4] Dipayan Sinha, Dr. Minal Moharir, Prof. Anitha Sandeep,

“Phishing Website URL Detection using Machine

Learning,” International Journal of Advanced Science and

Technology, vol. 29, no. 3, pp. 2495-2504, 2020.

[5] R. Kiruthiga, D. Akila, “Phishing Websites Detection

Using Machine Learning,” International Journal of Recent

Technology and Engineering (IJRTE), vol. 8, no. 2S11,

pp. 11-114, September 2019

[6] H. Alqahtani et al., "Cyber Intrusion Detection Using Machine Learning Classification Techniques," in Computing Science, Communication and Security, Singapore, 2020, pp. 121-131: Springer Singapore.

[7]Altyeb Altaher. Phishing websites classification using hybrid svm and knn approach. International Journal of Advanced Computer Science and Applications, 8(6):90–95, 2017. https://doi.org/10.14569/ijacsa.2017.080611

[8] Mahdieh Zabihimayvan and Derek Doran. Fuzzy rough set feature selection to enhance phishing attack detection. arXiv preprint arXiv:1903.05675, 2019. https://doi.org/10.1109/ fuzz-ieee.2019.8858884

[9]] Adwan Yasin and Abdelmunem Abuhasan. An intelligent classification model for phishing email detection. arXiv preprint arXiv:1608.02196, 2016. [17] Mohammad Almseidin, Maen Alzubi, Szilveszter Kovacs, and

[10]Mouhammd Alkasassbeh. Evaluation of machine learning algorithms for intrusion detection system. In 2017 IEEE 15th International Symposium on Intelligent Systems and Informatics (SISY), pages 000277–000282. IEEE, 2017. https://doi.org/10.1109/sisy.2017.8080566

[11] Nilima Khanna -J48 Classification (C4.5 Algorithm) in a Nutshell | by Nilima Khanna | Medium

[12] Mouhammad Alkasassbeh and Mohammad Almseidin. Machine learning methods for network intrusion detection. Icccnt 2018 - The 20TH International Conference On Computing, Communication And Networking Technologies, 2018

[13]Mouhammd Alkasassbeh, Ghazi Al-Naymat, AB Hassanat, and Mohammad Almseidin. Detecting distributed denial of service attacks using data mining techniques. International Journal of Advanced Computer Science and Applications, 7(1):436–445, 2016. https://doi. org/10.14569/ijacsa.2016.070159

14 Mouhammad Alkasassbeh. An empirical evaluation for the intrusion detection features based on machine learning and feature selection methods. Journal of Theoretical and Applied Information Technology, 95(22), 2017.

[15] Ryan J Urbanowicz, Melissa Meeker, William La Cava, Randal S Olson, and Jason H Moore. Relief-based feature selection: introduction and review. Journal of biomedical informatics, 2018. https://doi.org/10.1016/j.jbi.2018.07.014

[16] Alaa Tharwat. Classification assessment methods. Applied Computing and Informatics, 2018.